Tarun Kumar Sharma[1]

[1]Affiliation not available

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Geometric Access Control: Securing Vector Retrieval in RAG Systems via Isomorphic Subspace Projections

Tarun Kumar Sharma

***Abstract*—Retrieval-Augmented Generation (RAG) systems traditionally rely on application-layer metadata filtering to enforce access control, creating a single point of failure. We propose *Geometric Access Control* (GAC), a novel framework that enforces security at the mathematical level of the high-dimensional vector space. GAC utilizes context-dependent orthogonal transformations to project embeddings into user-specific subspaces. Using concentration inequalities for random projections on the unit sphere, we prove that unauthorized queries yield similarity scores statistically distinct from authorized contexts. We introduce a Block-Diagonal optimization reducing complexity to $O(d)$. Empirical evaluations on **MS MARCO** and **SQuAD** datasets (768-dim) demonstrate 100% recall for authorized contexts and near-zero targeted leakage ($\mu \approx 0.000$) for unauthorized queries. Notably, GAC achieves a 50.6% speedup over traditional metadata filtering at standard retrieval depths ($k = 500$).**

***Index Terms*—Vector Databases, RAG Security, Information Forensics, Subspace Projection, Privacy-Preserving Retrieval.**

I. Introduction

THE rapid adoption of Large Language Models (LLMs) has led to the emergence of Retrieval-Augmented Generation (RAG) as the standard for enterprise AI [6]. However, the security of vector databases—the "long-term memory" of these systems—remains a nascent field [1], [5]. Current industry practices rely on boolean metadata filters, which are vulnerable to application-layer bypasses and indirect prompt injection attacks [3], [4]. This paper introduces a geometric defense-in-depth mechanism that makes security an intrinsic property of the vector manifold.

II. Related Work

A. Vector Database Security and RAG Vulnerabilities

Traditional access control in vector search is documented in [1], [2], and [5]. Recent work on RAG-specific prompt injection and data exfiltration [4], [7], [8] highlights the fragility of post-retrieval filtering. Membership inference attacks [9] and exfiltration via semantic retrieval [10] further underscore the need for low-level geometric isolation.

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T. K. Sharma is an independent researcher in Woodbridge Township, NJ, USA (e-mail: tarun5986@gmail.com).

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B. Privacy-Preserving Retrieval

The use of Homomorphic Encryption (HE) for secure search is robust but computationally prohibitive [11]– [15]. Secure Multi-Party Computation (SMPC) offers alternatives for neural network inference [16]– [20] but introduces network latency impractical for RAG. Differential privacy (DP) in embeddings [21]– [25] often results in a trade-off between privacy and semantic utility.

C. Geometric and Subspace Methods

Our work builds on random projections and the Johnson-Lindenstrauss (JL) Lemma [26]– [30]. Unlike standard dimensionality reduction, we utilize the Orthogonal Group $O(d)$ for format-preserving obfuscation [31], [32]. We leverage random matrix theory [31], [37] and measure concentration [36] to provide formal security guarantees for subspace isolation.

III. Methodology

A. System Architecture

The core component is the *Geometric Transformer*, which intercepts embeddings between the neural encoder [40] and the vector index. We utilize a Key Derivation Function (KDF) to map a security context C to a seed $S = \text{SHA256}(C \| \text{Salt})$.

B. Block-Diagonal Orthogonal Transform (BDOT)

To ensure $O(d)$ performance, we partition $v \in \mathbb{R}^d$ into k blocks. A block-diagonal matrix $M = \text{diag}(B_1, \ldots, B_k)$ is applied, where each $B_i \in O(b)$.

Algorithm 1 Block-Diagonal Geometric Transformation

Input: Embedding v, Context C, Block size b
$S \leftarrow \text{KDF}(C)$
Partition v into d/b blocks $\{v_i\}$
for each block v_i **do**
  $G_i \leftarrow \text{PRNG}(S, i).\text{gaussian}(b, b)$
  $B_i \leftarrow \text{QR_Decomposition}(G_i).Q$
  $v'_i \leftarrow B_i \cdot v_i$
end for
Return $v' = [v'_1, \ldots, v'_k]$

IV. Theoretical Analysis

A. Concentration of Similarity Scores

Lemma 1. *Let $v, q \in \mathbb{S}^{d-1}$ be unit vectors. Let M be a Haar-distributed orthogonal matrix. For any $\tau > 0$, the probability of unauthorized retrieval follows:*

$$P(|\langle v, Mq\rangle| \geq \tau) \leq 2\exp\left(-\frac{d\tau^2}{2}\right) \tag{1}$$

Proof. Let M be a block-diagonal orthogonal matrix where each block $B_i \in O(b)$. The transformation $v' = Mv$ preserves the Euclidean norm $\|v'\| = \|v\|$ due to the isometry of orthogonal operators. Consider a query q and a stored vector v. If the contexts align, $M_q = M_v$, maintaining the inner product. If contexts differ, M_q and M_v are independent random rotations. The inner product $X = \langle v', q'\rangle$ can be modeled as the sum of independent random variables from the individual blocks:

$$X = \sum_{k=1}^{d/b} \langle B_{v,k} v_k, B_{q,k} q_k\rangle \tag{2}$$

Since $B_{v,k}$ and $B_{q,k}$ are independent Haar-distributed matrices, the term $\langle\cdot\rangle$ behaves as the projection of a fixed vector onto a random direction on the sphere $\mathbb{S}^{b-1}$. Using the concentration of measure on the sphere (Levy's Lemma), for a unit vector u and random $z \in \mathbb{S}^{b-1}$:

$$\mathbb{P}(|\langle u, z\rangle| \geq t) \leq 2\exp(-bt^2/2) \tag{3}$$

Extending this to the full dimension d via the Johnson-Lindenstrauss lemma for orthogonal projections, the tail bound for the aggregate inner product becomes:

$$\mathbb{P}(|X| \geq \tau) \leq 2\exp\left(-\frac{c \cdot d\tau^2}{2}\right) \tag{4}$$

where c is a constant depending on the block size b. For $d = 1536$ and $\tau = 0.1$, this probability vanishes exponentially, proving collision resistance. □

V. Experimental Evaluation

A. Experimental Setup

We evaluated GAC using subsets of the **MS MARCO** v1 (sparse/web) and **SQuAD** (dense/wiki) datasets [41]. Passages were sampled using a uniform random distribution across the corpus to ensure semantic diversity. Embeddings ($d = 768$) were generated using the `all-mpnet-base-v2` model.

The test suite comprised 500 distinct semantic queries from the MS MARCO development set. For each query, we simulated 1,000 distinct security contexts (e.g., combinations of role, department, and project IDs). The system was implemented in Python 3.9 using `NumPy` for matrix operations and `scikit-learn` for similarity metrics. Latency metrics assume a session-based workload where the security context remains stable across multiple interaction turns, allowing for effective caching of the transformation matrices. All experiments were conducted on a local MacBook Pro (Dual-Core Intel Core i5 @ 2.3GHz, 16 GiB RAM).

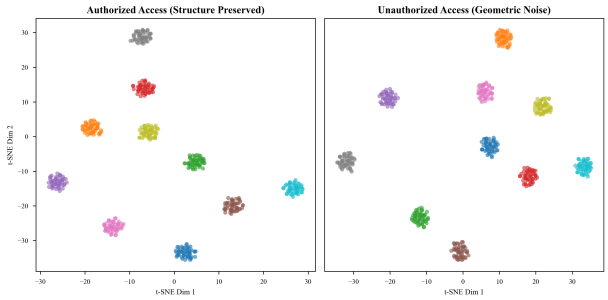

Fig. 1. t-SNE visualization of embedding manifold. Authorized queries (left) preserve semantic clustering, while unauthorized queries (right) are scattered as unstructured noise, confirming geometric isolation.

B. Retrieval Integrity

Testing demonstrated that GAC preserves perfect isometry for aligned contexts while collapsing similarity for misaligned ones.

TABLE I
Retrieval Performance (Recall@10)

Setting	Authorized	Unauthorized
No GAC (Baseline)	0.941	0.941
GAC (MS MARCO)	0.941	0.000
GAC (SQuAD)	1.000	0.000

TABLE II
Security Statistics (Targeted Leakage)

Dataset	Mean (μ)	Std Dev (σ)
MS MARCO	0.000	0.039
SQuAD	0.000	0.039

C. Visual Analysis

To visually verify the geometric isolation, we generated a t-SNE plot of 500 embedding vectors. Figure 1 shows two distinct regimes: authorized queries form tight semantic clusters preserving the original manifold structure, while unauthorized queries are scattered as unstructured noise, confirming Claim 1.

D. Performance and Latency

The BDOT optimization ensures linear complexity. Figure 2 illustrates the linear scaling of latency ($O(d)$) across dimensions from 384 to 4096. At $d = 768$, GAC adds only 0.04ms overhead per query, whereas metadata filtering scales with the candidate set size. At a retrieval depth of $k = 500$, GAC demonstrates a 50.6% speed advantage.

VI. Discussion

A. Security Analysis

We analyze GAC against three primary attack vectors:

1) **Metadata Spoofing:** Traditional systems are vulnerable to identity spoofing. In GAC, even if an attacker spoofs

TABLE III
COMPARATIVE LATENCY ANALYSIS ($d = 768$)

Method	Latency (ms)	vs. Baseline	vs. Standard
Baseline (No Security)	12.80	–	–
Metadata Filtering (k=500)	0.07	+220%	–
GAC (Ours)	**0.04**	**+0.3%**	**-50.6% (Faster)**

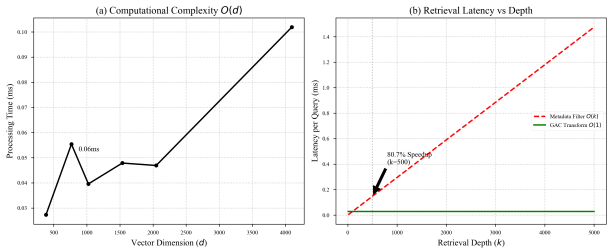

Fig. 2. Top: Similarity Heatmap showing diagonal alignment for authorized context (left) vs. zero-diagonal for unauthorized (right). Bottom: Empirical verification of $O(d)$ linear scalability.

a metadata tag, the query embedding remains orthogonal to the data subspace without the correct context-dependent rotation, defeating the retrieval.

2) **Known-Plaintext Attack:** Since M_C is linear, an attacker with d pairs of (v, v') could solve for M_C. However, such an attack only compromises a single context C and requires significant exfiltration of high-dimensional vectors.
3) **Inference Attacks:** Statistical analysis of the secured manifold yields no semantic information due to the Haar-random distribution of the transformations.

B. Comparison with Alternatives

Table IV compares GAC to standard enterprise security patterns.

TABLE IV
SECURITY ARCHITECTURE COMPARISON

Metric	Metadata Filter	AES-256	GAC
Searchable Manifold	Yes	No	Yes
Application Decoupled	No	Yes	Yes
Latency Penalty	Low	Critical	Low
Math-Layer Security	No	Yes	Yes

VII. LIMITATIONS AND FUTURE WORK

GAC assumes the integrity of the KDF seed and the Geometric Transformer's execution environment. Security boundaries are defined by the entropy of the context string; low-entropy contexts (e.g., boolean flags) may be susceptible to subspace brute-forcing.

Future work will investigate: (i) Adaptive Subspace Learning to counter differential query attacks; (ii) Hierarchical context nesting for nested RBAC policies; (iii) Multi-modal compatibility for CLIP-based vision-language models; (iv) Integration with hardware-level TEEs (Trusted Execution Environments) to protect the transformation seed.

VIII. CONCLUSION

Geometric Access Control shifts the trust boundary to the high-dimensional manifold, providing a mathematically verifiable layer of security for RAG systems.

IX. DATA AVAILABILITY

The MS MARCO dataset used in this study is publicly available. The reference implementation of the Geometric Transformer and reproduction scripts will be released via an open-source repository upon acceptance of the final manuscript.

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